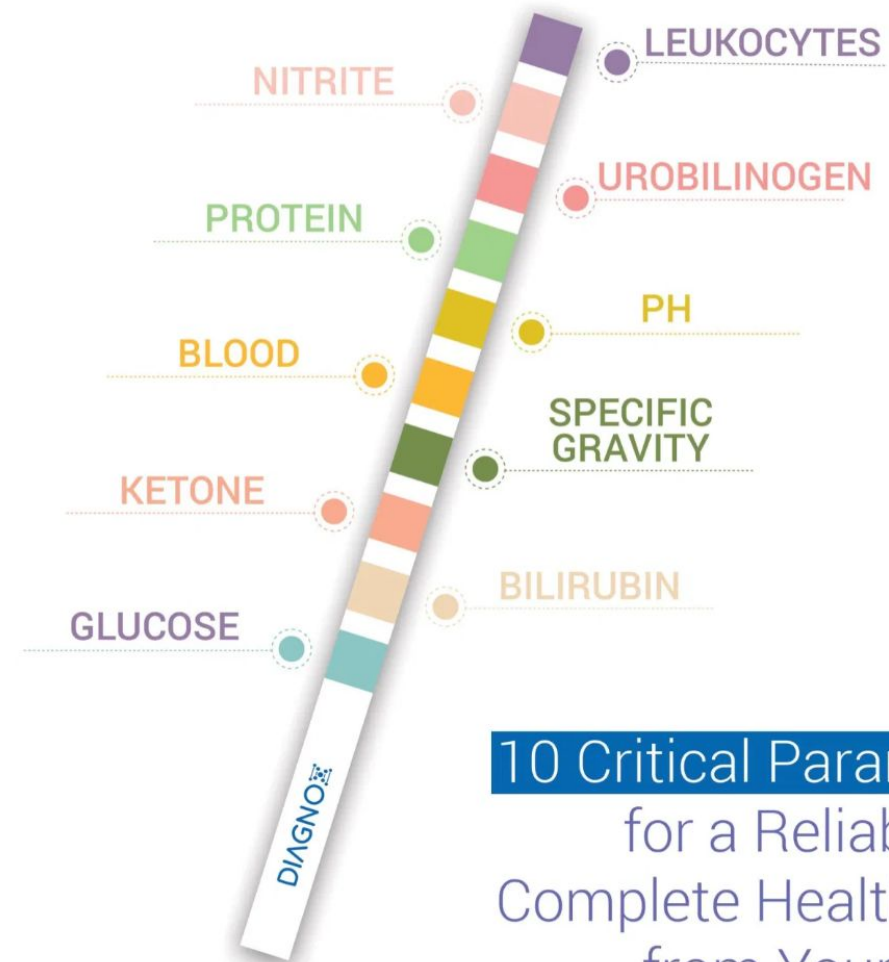


Urinalysis Automation using Smart Phone Cameras



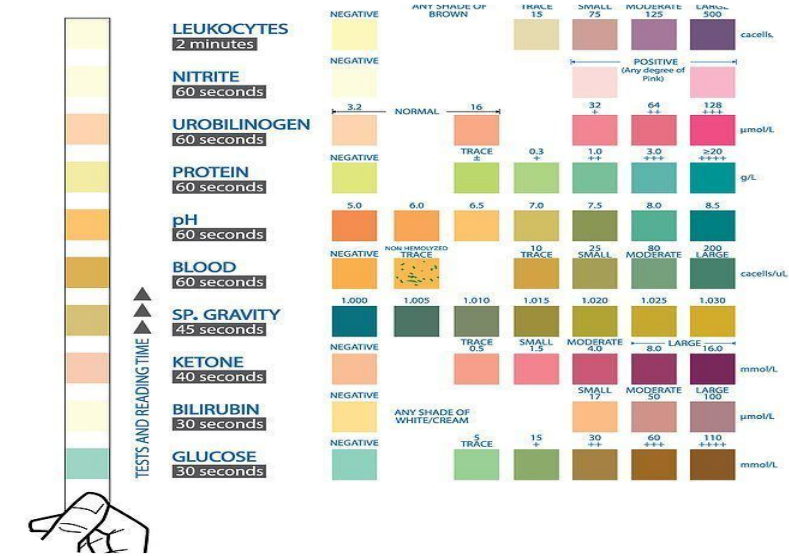
10 Critical Parameters
for a Reliable and
Complete Health Scan
from Your Home

Traditional Procedure



Extract test strip.

Immerse in specimen.



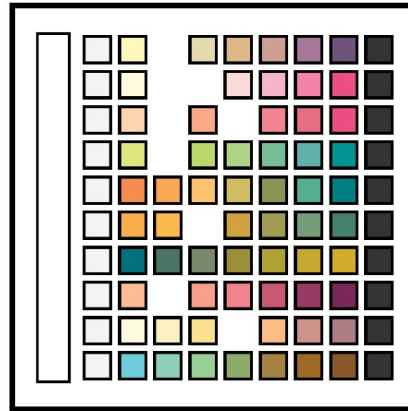
Manually match colors.

Overview of Automated Procedure

Step 1

Immerse
test strip.

Step 2



Position it
on
Calibration
Chart.

Step 3



Capture the
photograph
& view
results.

Algorithm Overview

1. Calibration Chart Design.

2. Image Segmentation to read captured colors.

3. Color Correction matrices development using multivariate linear regression.

4. Color comparison for semi-quantitative test report.

Pros of Implemented Algorithm

Accurate results under different lightening conditions.

Consistent performance across different smart phone cameras.

Economic & ubiquitous solution

Greater connectivity & medical record keeping.

Performance Analysis



Image Captured using Xiomi phone
under white room light.

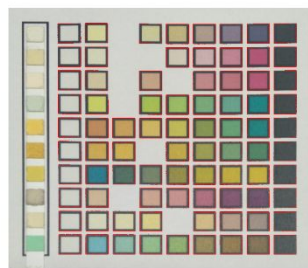
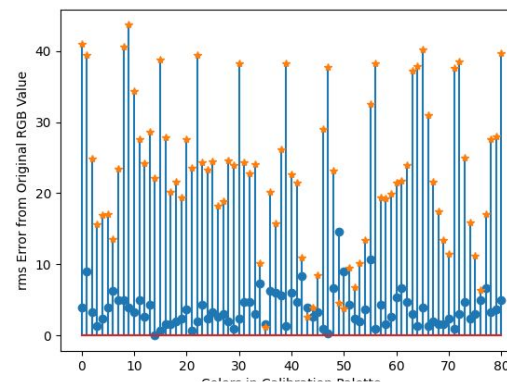
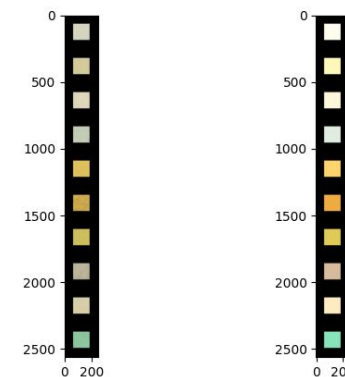


Image Straightening &
Segmentation.

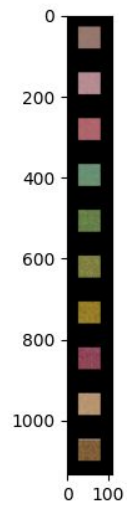


Error Analysis

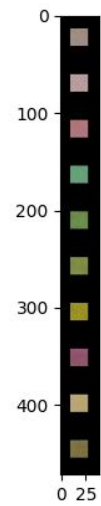
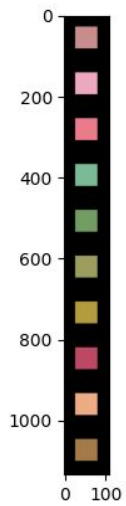


Output Analysis

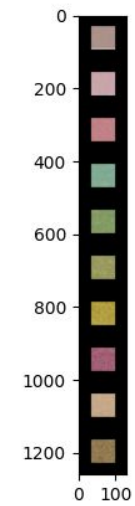
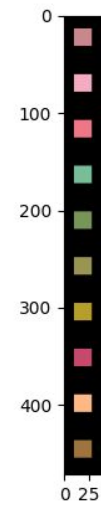
Performance across multiple Smartphones



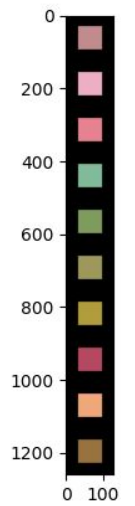
Xiomi Poco X3 Smartphone



Oppo A 57 Smartphone



Samsung S9 plus Smartphone



INSTRUCTIONS FOR USE

1. Tear open the blue pouch and obtain the urinalysis test strip.

2. Immerse the strip in urine sample (within a cup) and remove immediately.

3. Next, place it on paper towel to remove excess urine.

4. Now position the strip on the black background, next to the color chart.
Ensure that all color patches fit within the black space.

5. After set time, capture photograph of color chart and test strip in mild lightening.
Caution: Try not to cast shadow over the color chart or the test strip.

6. Obtain results!

Capture Image

PATIENT BIODATA

Name:

Height:

Age:

Weight:

MEDICAL HISTORY

TEST RESULTS

1. Leukocytes:

2. Nitrie:

3. Urobilinogen:

4. Protein:

5. pH:

6. Blood:

7. Sp. Gravity:

8. Ketone:

9. Bilirubin:

10. Glucose:

Smart Phone Application Design

Thank you!

